

Roll No.....

END SEMESTER EXAMINATION MAY 2025

Name of the Program: B. Tech. (CSE)
Name of the Course: Disaster Management

Semester: VIIIth Semester
Course Code: TDM-881

Time: Three Hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions among a, b, & c in each main question.
- (iii) Each sub question carries 10 marks.
- (iv) Each question carries 20 marks.
- (v) For programming question provide typical output.

Q.1		(2x10=20 Marks)	
a.)	What are cloudbursts? Discuss their causes, impacts, and mitigation strategies, especially in the context of Uttarakhand.		CO1
b.)	Discuss the phenomenon of forest fires. What are their natural and human causes? Analyze their effects on biodiversity and the environment, and suggest effective mitigation strategies.		CO2
c.)	Describe the diversion of river routes using the Kosi River as an example. What were the environmental and human consequences? How can such disasters be prevented in the future?		CO3
Q.2		(2x10=20 Marks)	
a.)	Explain the inter-relationship between disasters and development. How can unplanned development increase vulnerability to disasters? Give examples to support your answer.		CO1
b.)	What is the role of land use planning in disaster mitigation? Discuss how changes in land use patterns can increase or decrease disaster risks.		CO6
c.)	Explain the concept of climate change adaptation. How can it be integrated into disaster risk reduction strategies at the local and national levels?		CO6
Q.3		(2x10=20 Marks)	
a.)	Discuss the roles and responsibilities of different stakeholders such as the community, Panchayati Raj Institutions, urban local bodies, state and central governments in disaster mitigation. How can coordinated efforts improve disaster resilience?		CO5
b.)	Explain the three stages of disaster management: pre-disaster, emergency, and post-disaster. What are the key activities undertaken at each stage to reduce disaster impact?		CO5

c.)	Describe the role of forecasting, early warning systems, and information-education-communication (IEC) strategies in disaster preparedness. Provide examples from India.	CO5
Q.4 (2x10=20 Marks)		CO6
a.)	Analyze the importance of disaster-resistant house construction and population reduction in vulnerable areas as part of disaster mitigation strategies. How can these measures help save lives?	
b.)	Evaluate the roles and coordination among political leaders, administrative agencies, NGOs, and communities during the post-disaster phase. How can effective collaboration improve recovery and resilience?	
c.)	Explain the structure and functions of the National Disaster Management Authority (NDMA). How does NDMA coordinate with state and district authorities during a disaster?	CO4
Q.5 (2x10=20 Marks)		CO1
a.)	Write short notes of Disaster Management with suitable explanation as: (i) Man-Induced Disasters (ii) Disaster Management Act, 2005 (iii) Nuclear Explosions	
b.)	What are the roles and responsibilities of the National Disaster Response Force (NDRF)? Highlight the importance of NDRF in emergency response with examples.	
c.)	Describe the functions of the National Institute of Disaster Management (NIDM). How does NIDM contribute to capacity building and research in the field of disaster management?	CO2